Full Stack Development with MERN

Project Documentation format

1. Introduction :

• **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
• **Team Members:**

Name	Role
AISWARIYA K	Team Leader

2. Project Overview :

- **Purpose:** The project aims to automate the **standard laptop procurement process** using the ServiceNow platform. The system allows employees to submit a Standard Laptop request through the Service Catalog.

  After submission, the request is sent for approval. Once approved, **Flow Designer automatically creates a Catalog Task and assigns it to the Hardware assignment group**, reducing manual effort and improving request tracking.

- **Features:** Standard Laptop request through Service Catalog

  - Request validation and submission
  - Approval and rejection workflow
  - Automated Flow Designer trigger
  - Automatic Catalog Task creation
  - Automatic assignment to **Hardware** group
  - Laptop requirement details in the task
  - Request and task status tracking
  - Hardware task completion
  - Improved visibility of the procurement process

3. Architecture :

• **Frontend:** The project does not use a separate React frontend.

The user interface is provided by the **ServiceNow platform**, primarily through:

- Service Catalog
- Standard Laptop Catalog Item
- My Requests / Requested Items

- Approval interface
- Catalog Task interface

• **Backend:** The project does not use a separate Node.js or Express.js backend.

The application logic is implemented using **ServiceNow Flow Designer**.

Employee
↓
Service Catalog
↓
Standard Laptop Request
↓
Approval
↓
Approved
↓
Flow Designer
↓
Catalog Task Created
↓
Assignment Group: Hardware
↓
Hardware Processing
↓
Task Completed

•**Database:** The project uses the **ServiceNow platform's built-in database** to store:

- Requests
- Requested Items
- Approval records
- Catalog Tasks
- User information
- Task status
- Assignment information

No separate MongoDB database is used.

4. Setup Instructions :

• **Prerequisites:**
- Internet connection
- Web browser
- ServiceNow Developer Instance
- ServiceNow account

- Access to Service Catalog
- Flow Designer access
- Appropriate configuration/admin permissions

• **Installation:** Since this is a ServiceNow application, there is no traditional git clone, npm install, or MongoDB setup.

Steps

1. Log in to the ServiceNow Developer Instance.
2. Open **Service Catalog**.
3. Create/configure the **Standard Laptop** catalog item.
4. Configure the required laptop request fields.
5. Configure the approval process.
6. Open **Flow Designer**.
7. Create the automation flow.
8. Configure the flow trigger for an approved request.
9. Add an action to create a **Catalog Task**.
10. Configure the task details.
11. Set **Assignment Group = Hardware**.
12. Activate the flow.
13. Submit a Standard Laptop request to test the workflow.

5. Folder Structure :

Server:

Service Catalog
Used by employees to submit laptop requests.

Approval
Handles approval or rejection of the submitted request.

Flow Designer
Automates the creation of the Catalog Task after approval.

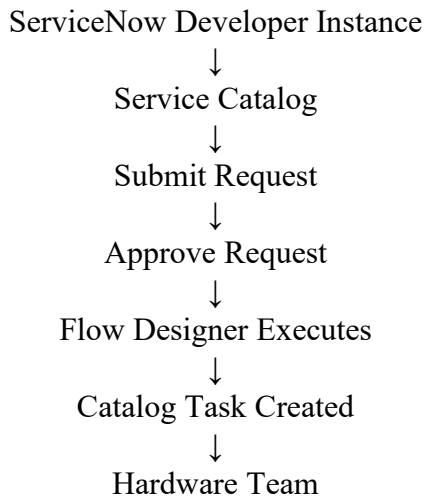
Catalog Task
Contains the laptop procurement/configuration task.

Hardware Assignment Group
Receives and processes the generated task.

6. Running the Application :

To Run

1. Log in to the ServiceNow Developer Instance.
2. Navigate to **Service Catalog**.
3. Select **Standard Laptop**.
4. Enter the required information.
5. Submit the request.
6. Open the submitted request.
7. Approve the request.
8. Flow Designer automatically executes.
9. Open the Requested Item.
10. Verify that a Catalog Task has been created.
11. Verify that the Assignment Group is **Hardware**.
12. Hardware team processes and completes the task.



7. API Documentation :

The project does **not expose custom REST APIs or backend endpoints**.

The automation is implemented using ServiceNow's native platform functionality.

Component	Purpose
Service Catalog	Submit laptop request
Requested Item	Store submitted request
Approval	Approve/reject request
Flow Designer	Automate workflow
Catalog Task	Create procurement/configuration task

Assignment Group Route task to Hardware

External API :

No external API is used.

8. Authentication :

Authentication is handled by the **ServiceNow platform**.

Users log in using their authorized ServiceNow credentials.

Authorization and access are controlled using ServiceNow's built-in:

- User roles
- Permissions
- Access Controls
- Assignment Groups

The project does not implement custom JWT authentication, sessions, or OAuth-based authentication.

9. User Interface :

Main UI Screens

1. Service Catalog
2. Standard Laptop Request Form
3. Submitted Request
4. Approval Screen
5. Requested Item
6. Catalog Task
7. Hardware Assignment
8. Task Completion / Status

TC-001 – Laptop Request Submission

Employee selects the Standard Laptop and submits the required information.

TC-002 – Request Approval

The approver reviews the submitted request and approves it.

TC-003 – Automatic Catalog Task

After approval, Flow Designer automatically creates the Catalog Task.

TC-004 – Task Details

The generated task contains the required laptop details and task description.

TC-005 – Hardware Assignment

The generated task is assigned to the **Hardware** group.

10. Testing :

Testing was performed on the ServiceNow workflow to verify that the complete procurement process functions correctly.

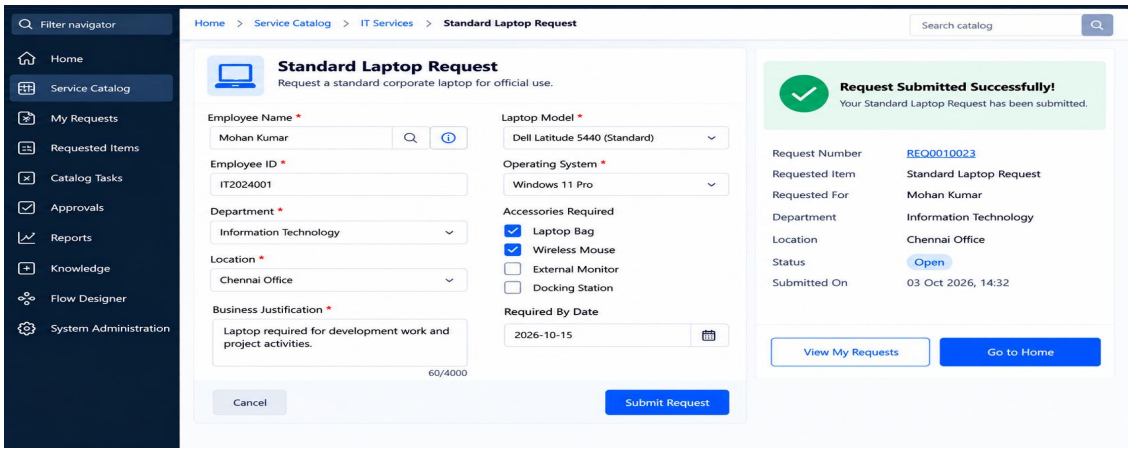
Testing Types

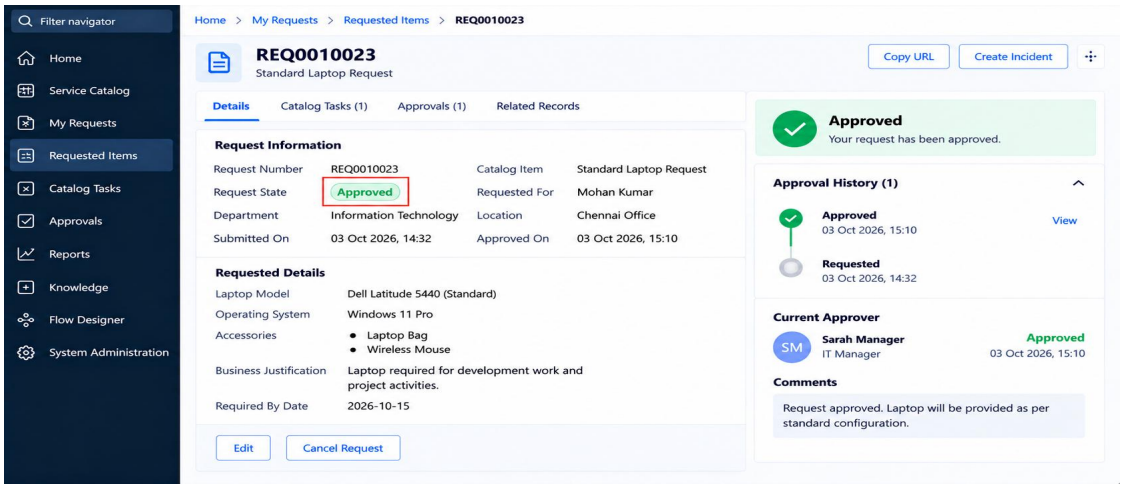
- Functional Testing
- Workflow Testing
- Approval Testing
- Automation Testing
- Assignment Testing
- Status Tracking Testing
- Performance Testing
- User Acceptance Testing

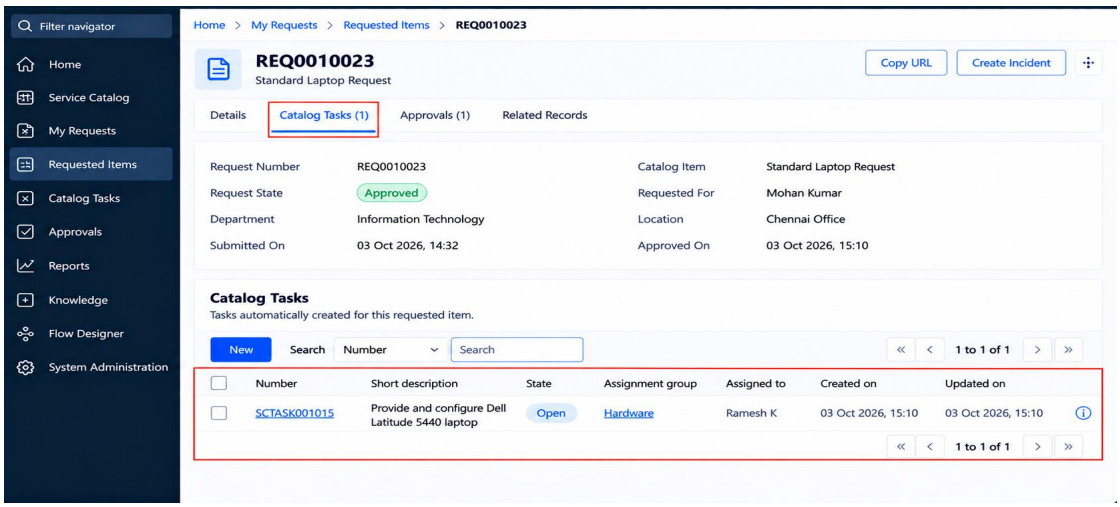
Functional Test Cases

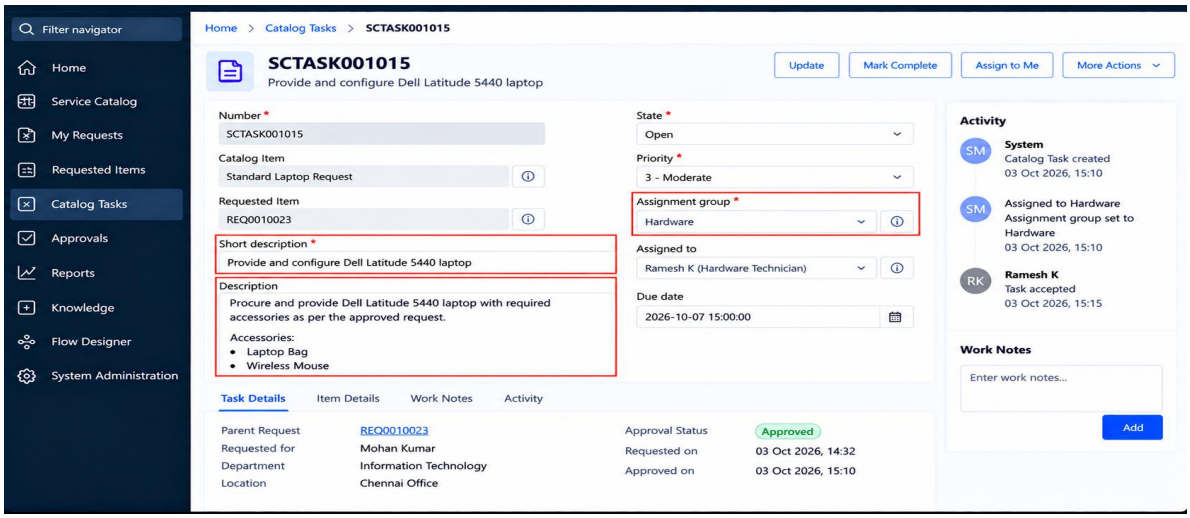
Test Case	Description	Expected Result	Status
TC-001	Submit Standard Laptop request	Request created	Pass
TC-002	Approve laptop request	Request approved	Pass
TC-003	Automatic Catalog Task creation	Task automatically created	Pass
TC-004	Verify task description	Required details populated	Pass
TC-005	Verify Hardware assignment	Task assigned to Hardware	Pass
TC-006	Reject laptop request	Request rejected	Pass
TC-007	Track request status	Status displayed correctly	Pass
TC-008	Complete Hardware task	Task completed	Pass

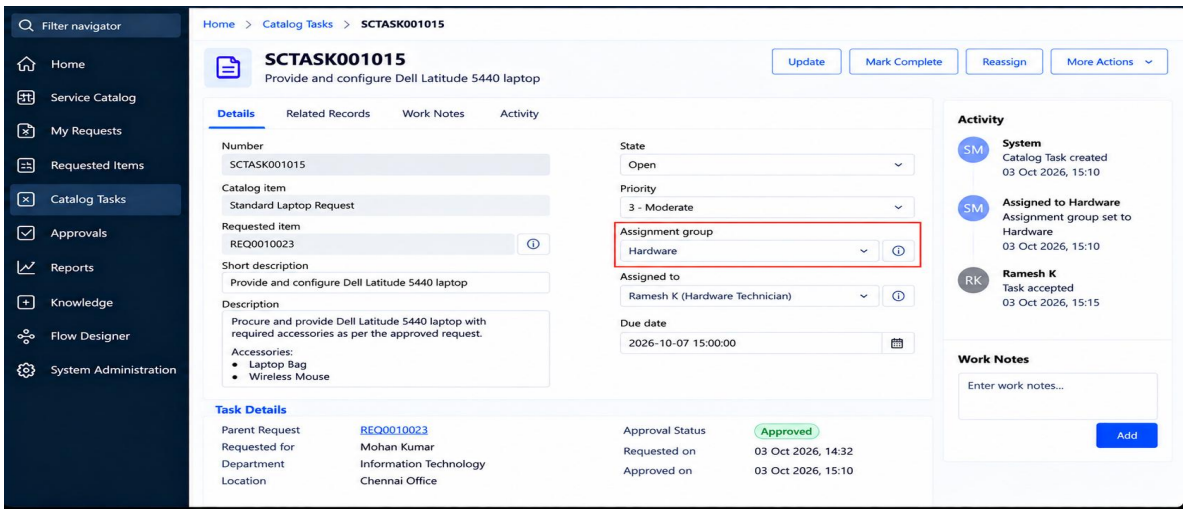
11. Screenshots or Demo :

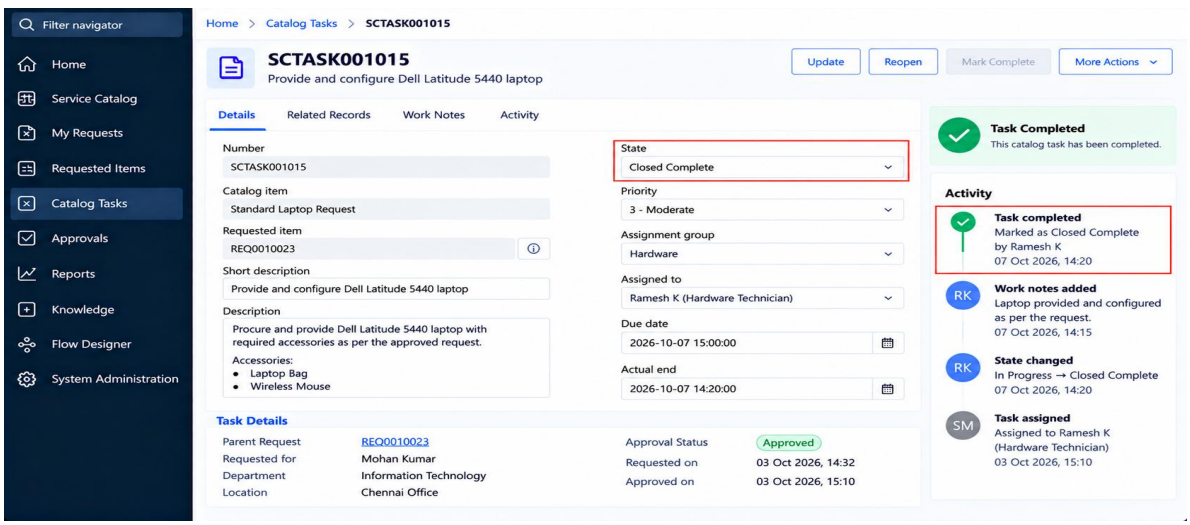

Filter navigator
Home
Service Catalog
My Requests
Requested Items
Catalog Tasks
Approvals
Reports
Knowledge
Flow Designer
System Administration
Home > Service Catalog > IT Services > Standard Laptop Request
Search catalog
Standard Laptop Request
Request a standard corporate laptop for official use.
Employee Name *
Mohan Kumar
Employee ID *
IT2024001
Department *
Information Technology
Location *
Chennai Office
Business Justification *
Laptop required for development work and project activities.
60/4000
Laptop Model *
Dell Latitude 5440 (Standard)
Operating System *
Windows 11 Pro
Accessories Required
Laptop Bag
Wireless Mouse
External Monitor
Docking Station
Required By Date
2026-10-15
Cancel
Submit Request
Request Submitted Successfully!
Your Standard Laptop Request has been submitted.
Request Number REQ0010023
Requested Item Standard Laptop Request
Requested For Mohan Kumar
Department Information Technology
Location Chennai Office
Status Open
Submitted On 03 Oct 2026, 14:32
View My Requests
Go to Home


Filter navigator
Home
Service Catalog
My Requests
Requested Items
Catalog Tasks
Approvals
Reports
Knowledge
Flow Designer
System Administration
Home > My Requests > Requested Items > REQ0010023
REQ0010023
Standard Laptop Request
Copy URL
Create Incident
Details
Catalog Tasks (1)
Approvals (1)
Related Records
Request Information
Request Number REQ0010023
Request State Approved
Department Information Technology
Submitted On 03 Oct 2026, 14:32
Catalog Item Standard Laptop Request
Requested For Mohan Kumar
Location Chennai Office
Approved On 03 Oct 2026, 15:10
Requested Details
Laptop Model Dell Latitude 5440 (Standard)
Operating System Windows 11 Pro
Accessories Laptop Bag, Wireless Mouse
Business Justification Laptop required for development work and project activities.
Required By Date 2026-10-15
Edit
Cancel Request
Approved
Your request has been approved.
Approval History (1)
Approved
03 Oct 2026, 15:10
View
Requested
03 Oct 2026, 14:32
Current Approver
SM
Sarah Manager
IT Manager
Approved
03 Oct 2026, 15:10
Comments
Request approved. Laptop will be provided as per standard configuration.


Filter navigator
Home
Service Catalog
My Requests
Requested Items
Catalog Tasks
Approvals
Reports
Knowledge
Flow Designer
System Administration
Home > My Requests > Requested Items > REQ0010023
REQ0010023
Standard Laptop Request
Copy URL
Create Incident
Details
Catalog Tasks (1)
Approvals (1)
Related Records
Request Number REQ0010023
Request State Approved
Department Information Technology
Submitted On 03 Oct 2026, 14:32
Catalog Item Standard Laptop Request
Requested For Mohan Kumar
Location Chennai Office
Approved On 03 Oct 2026, 15:10
Catalog Tasks
Tasks automatically created for this requested item.
New
Search
Number
Search
1 to 1 of 1
Number
Short description
State
Assignment group
Assigned to
Created on
Updated on
SCTASK001015
Provide and configure Dell Latitude 5440 laptop
Open
Hardware
Ramesh K
03 Oct 2026, 15:10
03 Oct 2026, 15:10
1 to 1 of 1







12. Known Issues :

- The workflow depends on correct ServiceNow configuration.
- Users require appropriate permissions to access the Service Catalog and workflow records.
- Flow execution may take a short amount of time after approval.
- Incorrectly configured approval conditions may prevent the automation from triggering.
- Incorrect assignment-group configuration can prevent the task from reaching the Hardware team.
- The project currently focuses on **Standard Laptop procurement** and does not cover all IT equipment categories.

13. Future Enhancements :

The project can be extended with:

1. **Multiple IT Equipment Categories**
   Add desktop, monitor, keyboard, mouse, printer and other equipment requests.
2. **Automated Notifications**
   Send email or ServiceNow notifications when a request is submitted, approved, rejected or completed.
3. **Request Dashboard**
   Provide an administrative dashboard showing pending, approved, rejected and completed requests.
4. **SLA Management**
   Add SLA tracking for procurement and hardware task completion.
5. **Automated Escalation**
   Escalate requests that remain pending beyond the defined time.
6. **Inventory Integration**
   Integrate laptop inventory information to check available devices.
7. **Approval Hierarchy**
   Add multiple approval levels for higher-value equipment.
8. **Additional Catalog Items**
   Extend the same Flow Designer approach to other IT procurement processes.